



COSC 341 Practice Midterm Exam (Answer)

60 Minutes

Name: _____

Student Number: _____

Total: _____

Common design principles: Visibility, Feedback, Constraints, Mapping, Consistency, Affordance, Simplicity, Matching, Minimizing memory load, Diagnose/Recover errors, Control and freedom, Flexibility, Provide Help

Names mentioned in the history lecture: Vannevar Bush, Douglas Engelbart, Ivan Sutherland, Alan Kay

Usability Goals: Effectiveness, Efficiency, Safety, Utility, Learnability, Memorability.

PART I – True and False

Circle either true (T) or false (F) for each of the following statements:

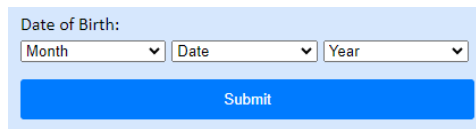
- 1) **T** F Interaction is a dialog with a cycle alternating between the user manipulating controls and the system responding with feedback
- 2) **T** F Internal consistency helps cultivate trust and communicates that a system is designed as a whole rather than a random collection of pieces
- 3) **T** F Typing an “i” instead of an “o” counts as a slip.
- 4) **T** F Images stored in the Drawables folder can be accessed programmatically using the `R.drawable.image_name` reference (here, `image_name` represents the image file name).
- 5) **T** F Persona gives a clear picture of the user's expectations and how they are likely to use an application
- 6) T **F** Vertical prototypes include the entire user interface with limited underlying functionalities

Part II – Multiple Choice

Circle only **one letter** for each question

1. A more _____ system reduces the time it takes to complete tasks as users spend more time with the system.
 - a. **Learnable**
 - b. Efficient
 - c. Effective
 - d. Safe
 - e. None of the above
2. Which of the following statements is CORRECT for margin and padding in the Android layout?
 - a. Padding and margin are two terms for the same concept in Android layouts
 - b. Padding is used to specify the distance between the View and its parent layout
 - c. Margin is used to specify the distance between sibling Views
 - d. Both B and C are correct
 - e. **None of the above**
3. People can remember _____ chunks of information.
 - a. Five $\pm$ two
 - b. Six $\pm$ two
 - c. **Seven $\pm$ two**
 - d. Eight $\pm$ two
 - e. None of the above

4. "Consider the interface below where users can enter their date of birth. By using dropdown menus, designers can limit the possible inputs to valid dates and prevent users from entering incorrect or invalid information." - Select one design principle from the list below that best meets the description.



- a. Affordance
 - b. Flexibility
 - c. Constraints**
 - d. Provide Help
 - e. None of the above
5. _____ conceptualized the Dynabook, a fundamental precursor to laptop and tablet computers
- a. Vannevar Bush
 - b. Douglas Engelbart
 - c. Ivan Sutherland
 - d. Alan Kay**
 - e. None of the above
6. One of the limitations of unstructured/open-ended interviews is:
- a. Not flexible to conduct
 - b. Time-consuming to conduct**
 - c. Answers from interviews suffer from a lack of detail
 - d. Do not provide the opportunity to probe for a deeper understanding
 - e. None of the above

Part III - Short Answer

1. As part of Lab 1, you developed a mobile application to facilitate user registration. Now, your objective is to evaluate the system's efficiency for completing a registration. Write two key metrics you would use to measure the application's efficiency.

Task completion time or Error rate

Answer: _____

2. Suppose you aim to gather user preferences for a mobile application and plan to use a rating system with a limited range of options. What type of scale would you use to capture users' preferences?

5-point Likert scales

Answer: _____

3. Provide an example of metaphors in user interface design and explain how they can enhance user interaction with interfaces.

Desktop, make Interaction language closer to users' own language, closer to task domain

4. In a _____, the presence of several relevant people together at the same time can encourage them to engage in a healthy discussion [Write the name of data gather technique]

Focus group

Answer: _____

5. What can a user **not** learn from a paper prototype apart from its visual appearance (i.e., look and feel)?

Response time

Answer: _____

6. When do we use the Wizard-of-Oz technique?

When it's difficult to prototype some piece of functionality

Answer: _____

7. Write one advantage of using questionnaires for data collection.

Allow researchers to carefully structure and formulate the data collection plan

Respondents can take these questionnaires at a convenient time and think about the answers at their own pace

Questionnaire can reach every corner of the globe

Advantage: _____

8. "Progress bars are commonly employed to show task status or file upload progress. They visually indicate completion by filling up with color, displaying the percentage completed or the time remaining. Users can interpret this visual representation to estimate the task's completion time." Select one design principle from the list below that best meets the description and write it in the answer.

Visibility, Constraints, **Mapping**, Simplicity, Minimizing memory load, Control and freedom, Flexibility

Answer: _____

9. In context of usability principles listed in page 1, what concepts is the following example violated (only write one that you think the most appropriate):

- a. A system showing the following error message:

502 Bad Gateway

The proxy server received an invalid response from an upstream server. Sorry for the inconvenience. Please report this message and include the following information to us. Thank you very much!

Heuristic: **Matching**

- a. A comments section displays the up and down arrows:

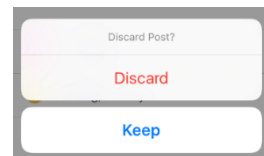


Heuristic: **Consistency**

10. Give an example of a prototyped interface in which the Wizard of Oz technique might be useful.

Answer: **Speech**

11. A social media application shows the following dialog box if a user taps on "Cancel" by mistake. The design principle _____ can be best illustrated with this example.



Answer: **Disgonosis/recover from error or Control and freedom**

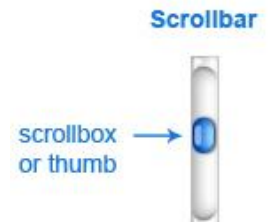
12. Give one example of a user interface that relies on recall over recognition.

Answer: **Conversational interface or command line**

13.] Besides time, what is one disadvantage of high-fidelity prototypes?

Answer: **Cost**

14. Consider the vertical scroll bar of a web browser or word processor. For each of the design principles below, give one way that the scrollbar uses the principle for effective design.



Mapping: The vertical motion of the thumb has a direct mapping to the motion of the document;

Visibility: The position of the thumb in the track indicates the scroll position; the thumb's size (relative to the track) indicates the amount of the document in view; the presence or absence of the scrollbar indicates whether scrolling is possible.

Feedback: When the thumb is dragged, the document scrolls immediately; when the scrollbar track is clicked, it flashes.

Constraints: Restrict users to move in the vertical direction